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ORIGINAL ARTICLE

Rapid and accurate initial alignment of the low-cost MEMS IMU chip dedicated for tilted RTK receiver

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Abstract

The new global navigation satellite system real-time kinematic (RTK) receivers with tilt compensation extend the applicability of high-precision RTK positioning to previously restrictive environments since the pole is no longer required to be maintained vertically. The rapid and accurate initial alignment of the built-in low-cost inertial measurement unit (IMU) is a critical technical issue and currently remains a challenge for the tilted RTK. We address this problem by proposing a method specific to the determination of the IMU initial heading for the tilted RTK. In this approach, the IMU heading is calculated from the angle between the inertial navigation system (INS)- and RTK-indicated position increment vectors in the horizontal plane based on the fact that the INS- and RTK-indicated trajectories in the short term are similar in shape. Instead of conventional INS mechanization, the INS-indicated trajectory is determined using the gyro-derived attitude and pole length by treating the pole movement as the motion of a rigid body with a fixed point to enhance the trajectory accuracy. The experiment results indicate that the initial heading is accurately determined to 1.15° at a 98.2% confidence level within only 2–3 s when using a dollar-level IMU chip without any additional sensors, e.g., a magnetometer. The proposed algorithm satisfactorily meets the requirements of the tilted RTK application, and its accuracy and convergence speed are higher than those of the current methods reported in publications and product sheets.

Keywords INS initial alignment · RTK with tilt compensation · GNSS surveying · Low-cost IMU · Rigid body motion

Introduction

Real-time kinematic (RTK) positioning is a global navigation satellite system (GNSS) navigation technique with a centimeter-level accuracy. In traditional RTK surveys, the pole on which the rover receiver is mounted should be vertically placed, as depicted by the dashed line in Fig. 1. Then, the antenna phase center (APC) position is reduced to the point of interest on the ground, by considering the vertical offset between the APC and pole tip.

In recent years, RTK receivers with a tilt compensation function, hereafter referred to as tilted RTK receivers, have emerged, which do not require the pole to be held vertically when measuring points on the ground, as illustrated in Fig. 1. Leica Geosystems AG (Switzerland), Trimble Inc. (Sunnyvale, California, USA), and several Chinese RTK receiver manufacturers have developed tilted RTK receivers. The tilted RTK technique makes surveying much more convenient and extends the applicability of high-precision RTK positioning to restrictive environments (Luo et al. 2018). For example, certain targets that were previously inaccessible by traditional RTK approaches, such as building corners and obstructed points, could be measured by tilting the pole. The position coordinates of the point to be surveyed are determined from the APC position through lever-arm correction using the pole attitude and lever-arm vector, which is a well-known issue in the field of multisensory integration. Determining the pole attitude based on a built-in inertial navigation system (INS) aided by RTK positions has become the mainstream in the latest generation of tilted RTK receivers. Figure 2 depicts the data processing scheme of an inertial measurement unit (IMU)-based tilted RTK receiver. To control the cost, only low-cost microelectromechanical system (MEMS) IMU chips are acceptable for tilted RTK receiver products. It is clear that the principle of tilt compensation

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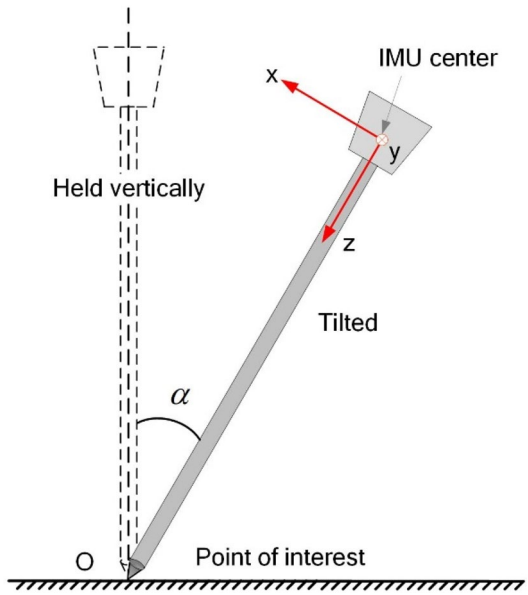

Fig. 1 Illustration of an RTK receiver with a tilt compensation function, and the definition of the IMU body frame (b-frame, lateral view). The IMU measurement center is assumed to coincide with the GNSS antenna phase center (APC) for analysis convenience

is straightforward, while the critical points for tilted RTKs involve the rapid initialization of the receiver attitude, i.e., the initial INS alignment, and maintenance of the attitude accuracy of the low-cost INS/RTK integrated solution at an acceptable level. We focus on addressing the former problem, i.e., the rapid and accurate INS initial alignment as labeled with the red color in Fig. 2, in particular, on the determination of the IMU initial heading.

INS initial alignment challenges arise for a tilted RTK receiver as commonly encountered in other situations using low-cost INSs, such as low-cost GNSS/INS integrated navigation. Alignment, i.e., attitude initialization, is the process whereby the orientation of the axes of an INS is determined with respect to the reference axis system (Titterton and Weston 2004), e.g., the process of calculating the initial roll, pitch and heading angles. For a tilted RTK receiver, it is necessary to accomplish accurate INS alignment within a very short period of time, e.g., a few seconds, to achieve a rapid reaction. As analyzed in the appendix, the roll and pitch angles could be initialized quickly to sufficient accuracy even for a low-cost chip-level IMU. Hence, it is the heading initialization that poses the major difficulty in initial alignment, where a 1.15–2° accuracy is typically required to be achieved, as derived in the appendix.

The issue of INS alignment has been studied in numerous studies (Groves 2013; Titterton and Weston 2004; Wu and Pan 2013), while this problem for low-cost INSs has been less frequently addressed. Traditionally, there are several types of fundamental alignment approaches: 1) self-alignment using gyrocompassing where the INS remains stationary with respect to the earth; 2) direct initialization from other sensors or systems; and 3) in-motion alignment with other aiding through data fusion. Self-alignment on a fixed platform requires high-precision aviation-grade gyros or better (Groves 2013). Thus, it is obviously not feasible for low-cost IMU chips in tilted RTK receivers. The heading could also be obtained directly from another sensor, such as an electronic magnetometer, a high-grade master reference (Titterton and Weston 2004), and dual- or multiple-antenna GNSS equipment (Chang et al. 2017; Dai et al. 2009). However, the accuracy of a magnetometer is sensitive to ferrous metals and has been proven to be not robust in the RTK survey environment (Luo et al. 2018), and the use of a dual-antenna GNSS master reference is generally expensive and not practical for portable tilted RTK receivers. In case of INS is in motion, GNSS velocity matching is an effective approach to coarsely determine the initial heading by $\psi = \tan^{-1}(v_E/v_N)$, where v_E and v_N are the east

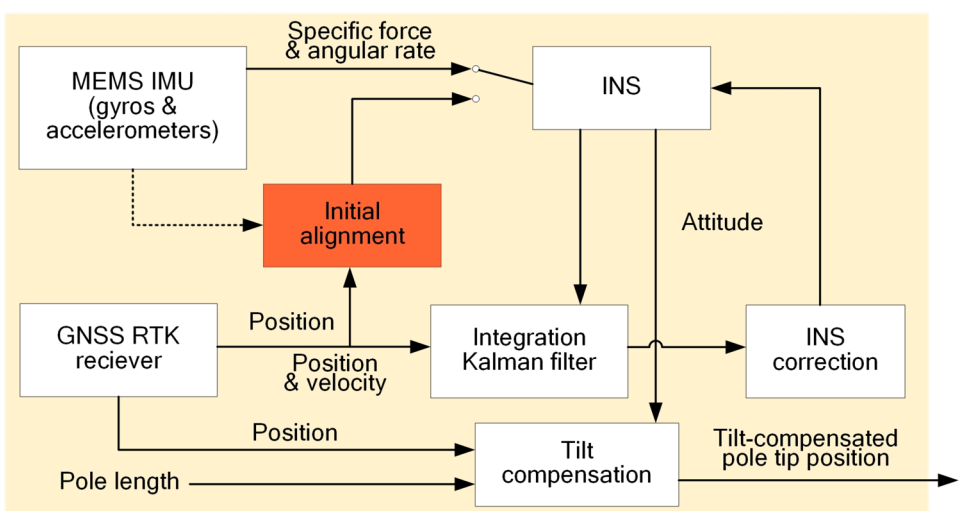

Fig. 2 Basic data processing schematic of the IMU-based tilted RTK receiver

and north velocity, respectively, measured by the GNSS. With this approach, the alignment error reaches 6° as demonstrated by Shin and El-Sheimy (2004), and the accuracy could be improved with GNSS aiding in the fine alignment stage (Rogers 2007). The alignment may converge to a 1° accuracy in 50 s or longer, and the host vehicle must perform notable maneuvering to accelerate the alignment process and increase the accuracy achieved. These in-motion alignment methods could be exploited in low-cost INS/GNSS integrated systems for land vehicles or unmanned aerial vehicles (UAVs). However, an alignment period exceeding 50 s is too long to be accepted for tilted RTK positioning applications, and it becomes impractical to perform the required maneuvering.

Limited information on tilted RTK initialization can also be found in technical documents provided by manufacturers. The Leica GS18 T instrument from Leica Geosystems AG, Switzerland, is reported to achieve an initial 2°-attitude accuracy within seconds through several meters of movement (Luo et al. 2018). The iRTK5X tilted RTK receiver from Hi-Target Surveying Instrument Co., Ltd., Guangzhou, China, is reported to achieve rapid initialization by using a high-grade IMU (http://en.hi-target.com.cn/product/prodetail.aspx?pid=201). The inertial-based tilted RTK receivers X12 and i90 from Shanghai Huace Navigation Technology Ltd., Shanghai, China, require a movement of approximately 15 to 30 m and dozens of seconds to complete initialization and INS heading convergence (https://www.chcnav.com/product-detail/i90-imu+-+rtk-gnss). However, technical details on the initial alignment method used by these commercial instruments are not available.

We summarize the literature review as follows: (1) There is little research that addresses the INS initial alignment problem specific for the newly emerged tilted RTK. (2) The conventional INS alignment methods designed for aviation and land vehicle applications do not perfectly meet the requirements for tilted RTK applications. (3) There remains an urgent need for an improved method of initial heading determination for tilted RTK receivers in the industrial community. Preferably, this method should be quick and highly accurate. We propose a rapid and accurate INS alignment method specific for tilted RTK receivers in response to the above technical challenges and application demands. In the proposed method, the IMU heading is calculated as the angle between the INS- and RTK-indicated APC position increment vectors in the horizontal direction based on the fact that the INS- and RTK-indicated trajectories are similar in shape. Instead of conventional INS mechanization, the INS-indicated trajectory is determined using the gyro-derived attitude and pole length by treating the pole movement as the motion of a rigid body with a fixed point to enhance the trajectory accuracy. The experimental results show that the initial heading could be determined at a 1.15° accuracy in only 2 s, which is more accurate and time-efficient than current methods and satisfies the requirements of tilted RTK applications.

The principle of the proposed method is derived straightforwardly, and the practical implementation of this new algorithm is described in detail. Thereafter, the experimental results are analyzed and discussed to verify the proposed method.

Methodology

The proposed INS alignment method involves a specific motion, as illustrated in the left panel of Fig. 3: the pole tip is fixed at a ground point O, and the pole is then tilted to an arbitrary position A, after which it is shifted to another position B to finish the alignment motion. IMU and RTK measurements are simultaneously recorded. Such a simple but specific operation benefits the algorithm design greatly by regarding the pole movement as a rigid body motion with

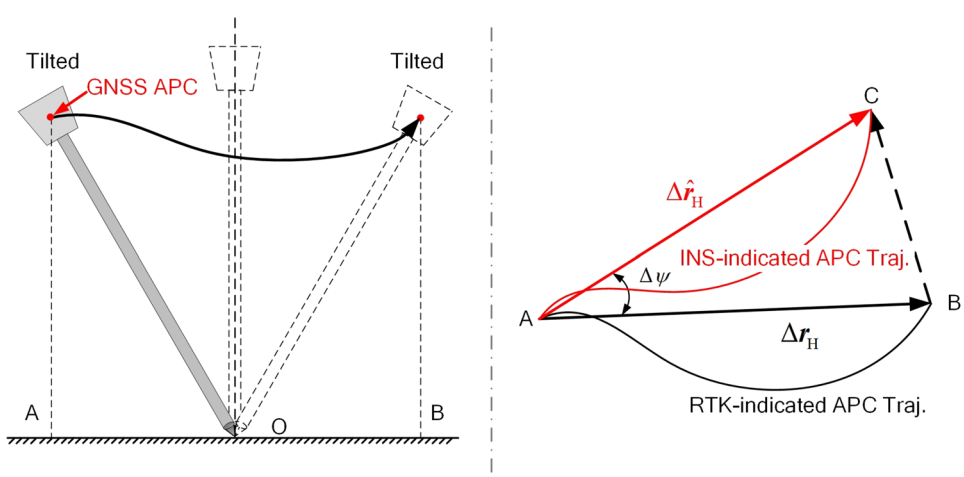

Fig. 3 Illustration of the proposed INS alignment principle for tilted RTK receivers. The alignment involves shifting the pole from tilt position A to tilt position B. The INS- and RTK-indicated trajectories are similar in shape in the horizontal plane (top view) due to the existence of $\Delta\psi$

a fixed point. The corresponding data processing is characterized as follows:

(1) It is assumed that the initial heading angle of the IMU is zero. Then, the INS-indicated APC trajectory, i.e., the position history, is calculated using the gyro-derived attitude and pole length by treating the pole movement as the motion of a rigid body with a fixed point to enhance the trajectory accuracy.
(2) The RTK-indicated APC trajectory, i.e., the discrete position history, is obtained from the RTK receiver.
(3) The offset of the assumed initial heading angle in step 1, denoted by $\Delta\psi$, is then determined as the angle between the INS- and RTK-indicated APC position increment vectors in the horizontal direction based on the fact that the INS-indicated and RTK-indicated trajectories are similar in shape, as depicted in the right panel of Fig. 3.

Principle

The derivation and practical implementation of the proposed algorithm are introduced below. The pole tip position is related to the APC position by considering lever-arm correction, as follows:

$$\boldsymbol{r}_I^n(t) = \boldsymbol{r}_T^n - \mathbf{C}_b^n(t)\boldsymbol{l}^b \tag{1}$$

where r_I^n and r_T^n are the positions of the IMU measurement center and pole tip, respectively, resolved in the local geographic north-east-down frame, i.e., the n-frame, and $\boldsymbol{l}^b$ is the level arm vector from the IMU measurement center to the pole tip in the IMU sensor frame, i.e., the forward-right-down b-frame, as depicted in Fig. 1. It should be noted that the IMU measurement center is assumed to coincide with the APC for simplicity. The above equation expresses the basic tilt compensation principle of a tilted RTK receiver, which is actually a lever-arm correction issue.

We assume that the IMU is aligned with the pole in attitude or that the misalignment angles between the b-frame and pole frame have been accurately calibrated. Thus, l^b has a nonzero value l for the third element (the z-component), and the other elements are zeros, i.e., $\boldsymbol{l}^b = \begin{bmatrix} 0 & 0 & l \end{bmatrix}^T$, where l denotes the length between the APC and pole tip. $\mathbf{C}_b^n$ denotes the direction cosine matrix that transforms vectors from the b-frame to the n-frame. The IMU attitude is defined as follows: The n-frame is rotated at successive rotation angles ψ, θ, and ϕ about the z-axis, intermediate y'-axis, and tertiary x"-axis to finally become aligned with the b-frame. The attitude matrix in terms of the Euler angles is given by equation (24) in the appendix. When shifting the pole, $\boldsymbol{r}_I^n$ and $\mathbf{C}_b^n$ change over time, denoted by t, due to the alignment movement. In addition, $\boldsymbol{r}_T^n$ and l^b are constant vectors, and $\boldsymbol{r}_T^n$ is determined with the RTK method by maintaining the pole vertically, and $\boldsymbol{l}^b$ can be accurately measured in the laboratory. At the start time epoch t_0, we have:

$$\boldsymbol{r}_I^n(t_0) = \boldsymbol{r}_T^n - \mathbf{C}_b^n(t_0)\boldsymbol{l}^b \tag{2}$$

The APC position increment vector from t_0 to t, denoted by $\Delta\boldsymbol{r}_I^n(t)$, is then defined as:

$$\begin{aligned}\Delta\boldsymbol{r}_I^n(t) &= \boldsymbol{r}_I^n(t) - \boldsymbol{r}_I^n(t_0)\\ &= \mathbf{C}_b^n(t_0)\boldsymbol{l}^b - \mathbf{C}_b^n(t)\boldsymbol{l}^b\end{aligned} \tag{3}$$

The general updating algorithm for $\mathbf{C}_b^n(t)$ is constructed as follows using the direction cosine matrix product chain rule (Wu and Pan 2013):

$$C_b^n(t) = C_{b(t)}^{n(t)} = C_{n(0)}^{n(t)} C_{b(0)}^{n(0)} C_{b(t)}^{b(0)} \tag{4}$$

where $\mathbf{C}_{b(t)}^{b(0)}$ and $\mathbf{C}_{n(0)}^{n(t)}$ account for the attitude changes of the b- and n-frame, respectively, between times t_0 and t. $\mathbf{C}_{b(0)}^{n(0)}$ is the constant initial attitude matrix to be determined during alignment (Wu and Pan 2013). Since we shift the pole top only within several meters and accomplish the alignment in about 2 s, the change in the n-frame orientation is negligibly small, e.g., smaller than $15/3600 \times 2 \approx 0.008°$, where 15°/h is the earth rotation rate. Thus, we obtain:

$$\mathbf{C}_b^n(t) = \mathbf{C}_{b(t)}^n \approx \mathbf{C}_{b(0)}^n \mathbf{C}_{b(t)}^{b(0)} \tag{5}$$

Equation (5) updates the attitude matrix $\mathbf{C}_b^n$ based on $\mathbf{C}_{b(t)}^{b(0)}$ to account for the angular rate of the b-frame to the nonrotating space (Savage 1998). $\mathbf{C}_{b(t)}^{b(0)}$ is calculated by integrating the gyro measurements. The errors contained in the computed matrix $\hat{\mathbf{C}}_b^n(t)$ originate from both the error in $\hat{\mathbf{C}}_{b(0)}^n$ and the computed $\hat{\mathbf{C}}_{b(t)}^{b(0)}$. Thus, we have

$$\hat{\mathbf{C}}_b^n(t) = \hat{\mathbf{C}}_{b(0)}^n \hat{\mathbf{C}}_{b(t)}^{b(0)} \tag{6}$$

The errors in the computed matrix $\hat{\mathbf{C}}_{b(t)}^{b(0)}$ are due to the gyro measurement errors, including the biases, scale factor error, cross-axis error, and random noise. In the following subsection, the error in $\hat{\mathbf{C}}_{b(t)}^{b(0)}$ will be proven sufficiently smaller than the error in $\hat{\mathbf{C}}_{b(0)}^n$.

In contrast, $\hat{\mathbf{C}}_{b(0)}^n$ may contain a large error. In practice, the INS heading is initialized by entering an arbitrary number, for example 0°, to start the navigator of a tilted RTK receiver. The entered initial heading may contain a large error, denoted as $\Delta\psi$, and we denote the corresponding initial attitude matrix containing this error as $\hat{\mathbf{C}}_{b(0)}^n$. We can assign the angular misalignment error to an error in the n-frame relative to the b(0)-frame without a loss of generality as $\hat{\mathbf{C}}_{b(0)}^n = \mathbf{C}_{b(0)}^{\hat{n}}$ (Savage 2007). With the use of the chain rule, we obtain:

$$\hat{\mathbf{C}}_{b(0)}^{n} = \mathbf{C}_{b(0)}^{\hat{n}} = \mathbf{C}_{n}^{\hat{n}}\mathbf{C}_{b(0)}^{n} \tag{7}$$

where $\mathbf{C}_{n}^{\hat{n}}$ is the direction cosine matrix that transforms vectors from the error-free n-frame to the misaligned $\hat{n}$-frame. The errors contained in $\mathbf{C}_{n}^{\hat{n}}$ stem from the errors in the initial Euler angles. As explained in the appendix, even for a MEMS IMU, the roll and pitch angles could be determined within a short static period through the accelerometer leveling procedure, whose error is notably smaller than that of the arbitrarily entered initial heading. Therefore, the error in the constant matrix $\hat{\mathbf{C}}_{b(0)}^{n}$ during the subsequent in-motion alignment is primarily the initial heading error. We then obtain:

$$\mathbf{C}_{n}^{\hat{n}} = \begin{bmatrix} \cos\Delta\psi & -\sin\Delta\psi & 0 \\ \sin\Delta\psi & \cos\Delta\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{8}$$

According to the analysis above, it is obvious that the attitude drift in $\hat{\mathbf{C}}_{b(t)}^{b(0)}$ is much smaller than the large error of the entered initial heading and the attitude error tolerance, e.g., 1.15–2°. Thus, Eq. (6) can be approximated as:

$$\hat{\mathbf{C}}_{b}^{n}(t) = \mathbf{C}_{n}^{\hat{n}}\mathbf{C}_{b(0)}^{n}\mathbf{C}_{b(t)}^{b(0)} \tag{9}$$

Considering the attitude errors, the computed APC position increment vector in (3), i.e., the INS-indicated APC position increment vector, can be represented as

$$\Delta\hat{\boldsymbol{r}}_{I}^{n}(t) = \hat{\mathbf{C}}_{b(0)}^{n}\left(\mathbf{I} - \mathbf{C}_{b(t)}^{b(0)}\right)\boldsymbol{l}^{b} \tag{10}$$

where $\Delta\hat{\boldsymbol{r}}_{I}^{n}(t)$ is the computed quantity of $\Delta\boldsymbol{r}_{I}^{n}(t)$ and $\mathbf{I}$ is the 3-by-3 identity matrix. Substitute (7) into (10) and rearrange, yield:

$$\begin{aligned}\Delta\hat{\boldsymbol{r}}_{I}^{n}(t) &= \mathbf{C}_{n}^{\hat{n}}\mathbf{C}_{b(0)}^{n}\left(\mathbf{I} - \mathbf{C}_{b(t)}^{b(0)}\right)\boldsymbol{l}^{b} \\ &= \mathbf{C}_{n}^{\hat{n}}\Delta\boldsymbol{r}_{I}^{n}(t)\end{aligned} \tag{11}$$

Thus, Eq. (11) implies that the INS-indicated APC trajectory is similar to the true trajectory in shape but with a constant rotation due to the occurrence of the initial heading bias. By substituting (8) into (11) and retaining the first two components, i.e., the horizontal coordinate components, we have:

$$\begin{bmatrix} \Delta\hat{r}_{I,N}^{n} \\ \Delta\hat{r}_{I,E}^{n} \end{bmatrix}_{t} = \begin{bmatrix} \cos\Delta\psi & -\sin\Delta\psi \\ \sin\Delta\psi & \cos\Delta\psi \end{bmatrix}\begin{bmatrix} \Delta r_{I,N}^{n} \\ \Delta r_{I,E}^{n} \end{bmatrix}_{t} \tag{12}$$

We denote $\Delta\hat{\boldsymbol{r}}_{H}^{n} = \left[\,\Delta\hat{r}_{I,N}^{n}\ \ \Delta\hat{r}_{I,E}^{n}\,\right]^{T}$ and $\Delta\boldsymbol{r}_{H}^{n} = \left[\,\Delta r_{I,N}^{n}\ \ \Delta r_{I,E}^{n}\,\right]^{T}$ as the INS-indicated and true horizontal position incremental vectors, respectively, at arbitrary time epoch t. $\Delta\hat{r}_{I,N}^{n}$ and $\Delta\hat{r}_{I,E}^{n}$ are the north and east components, respectively, of the computed incremental position. $\Delta r_{I,N}^{n}$ and $\Delta r_{I,E}^{n}$ denote the error-free north and east component of the incremental position, respectively. Equation (12) explicitly implies that vector $\Delta\hat{\boldsymbol{r}}_{H}^{n}$ is related to $\Delta\boldsymbol{r}_{H}^{n}$ by rotating over an angle of $\Delta\psi$, as illustrated in Fig. 3. This implies that once the INS-indicated and reference true trajectories are obtained, the initial heading bias $\Delta\psi$ could be calculated from the direction cosine value between the two planar vectors of $\Delta\boldsymbol{r}_{H}^{n}$ and $\Delta\hat{\boldsymbol{r}}_{H}^{n}$ as:

$$\cos\Delta\psi = \frac{\Delta\hat{\boldsymbol{r}}_{H}^{n}\cdot\Delta\boldsymbol{r}_{H}^{n}}{\left|\Delta\hat{\boldsymbol{r}}_{H}^{n}\right|\left|\Delta\boldsymbol{r}_{H}^{n}\right|} \tag{13}$$

where $\left|\Delta\hat{\boldsymbol{r}}_{H}^{n}\right|$ and $\left|\Delta\boldsymbol{r}_{H}^{n}\right|$ are the lengths of vectors $\Delta\hat{\boldsymbol{r}}_{H}^{n}$ and $\Delta\boldsymbol{r}_{H}^{n}$, respectively. The above equation is the basic principle of the determination of the initial heading of the proposed alignment method.

Practical implementation and accuracy analysis

It should be noted that the heading can be determined using only one sample of the simultaneous observation vectors $\Delta\hat{\boldsymbol{r}}_{H}^{n}$ and $\Delta\boldsymbol{r}_{H}^{n}$, as expressed by (12) and (13). In practice, the reference trajectory is computed from the RTK positions. In this case, only two epochs of INS- and RTK-indicated position measurements are needed, and the initial heading could be determined within 2 or 3 s at a GNSS RTK sampling rate of 1 Hz. If multiple measurement samples are incorporated in the alignment calculation, the solution of the initial heading could be obtained with the least square (LS) estimator, and the corresponding estimation accuracy could also be evaluated by least squares. More details on the LS estimation can be found in Leick et al. (2015). Alignment using multiple measurements may enhance the accuracy to a certain extent, but this also leads to a longer alignment period. We choose to preliminarily assess the alignment accuracy based on a single measurement to demonstrate the feasibility of the proposed method.

Therefore, the initial heading estimation error, denoted by $\delta\hat{\psi}$, is approximately evaluated from Fig. 3, as

$$\delta\hat{\psi} = \delta\left(\Delta\hat{\boldsymbol{r}}_{H}^{n} - \Delta\boldsymbol{r}_{H}^{n}\right)/\left|\Delta\boldsymbol{r}_{H}^{n}\right| \tag{14}$$

where $\delta\left(\Delta\hat{\boldsymbol{r}}_{H}^{n} - \Delta\boldsymbol{r}_{H}^{n}\right)$ denotes the error of horizontal vector $\boldsymbol{r}_{BC}$, as shown in Fig. 3, and $\left|\Delta\boldsymbol{r}_{H}^{n}\right|$ is the length of the true horizontal position increment vector $\boldsymbol{r}_{AB}$. In practice, $\Delta\hat{\boldsymbol{r}}_{H}^{n}$ and $\Delta\boldsymbol{r}_{H}^{n}$ are computed from the INS- and the RTK-indicated APC trajectories, respectively. Therefore, the initial heading estimation accuracy is influenced by the error in the INS- and RTK-indicated position increment measurements and the length of the true position increment. For a given length $\left|\Delta\boldsymbol{r}_{H}^{n}\right|$, the more accurate $\Delta\hat{\boldsymbol{r}}_{H}^{n}$ and $\Delta\boldsymbol{r}_{H}^{n}$ are, the more accurate the initial heading should be. In contrast, the estimation error

is inversely proportional to the trajectory length. Considering the RTK position errors, Eq. (15) can be written as

$$\begin{aligned}\delta\hat{\psi} &= \delta\left(\Delta\hat{\boldsymbol{r}}_H^n - \Delta\tilde{\boldsymbol{r}}_H^n\right)/\left|\Delta\boldsymbol{r}_H^n\right| \\ &= \left(\delta\Delta\hat{\boldsymbol{r}}_H^n - \delta\Delta\tilde{\boldsymbol{r}}_H^n\right)/\left|\Delta\boldsymbol{r}_H^n\right|\end{aligned} \tag{15}$$

where the subscript tilde in $\Delta\tilde{\boldsymbol{r}}_H^n$ indicates that it is computed from the delta RTK positions which also contains errors. Neglecting the correlations between the errors in $\Delta\hat{\boldsymbol{r}}_H^n$ and $\Delta\tilde{\boldsymbol{r}}_H^n$, the heading alignment error is computed according to the error propagation law as:

$$\sigma_{\hat{\psi}} = \sqrt{\sigma_{\psi,I}^2 + \sigma_{\psi,G}^2} \tag{16}$$

where $\sigma_{\hat{\psi}}$ denotes the root mean square (RMS) error of the heading alignment results, $\sigma_{\psi,I}$ is the RMS of the INS-related heading alignment error (the RMS of $\delta\Delta\hat{\boldsymbol{r}}_H^n\Big/\left|\Delta\boldsymbol{r}_H^n\right|$), and $\sigma_{\psi,G}$ is the root variance of the RTK-related heading alignment error (the RMS of $\delta\Delta\tilde{\boldsymbol{r}}_H^n\Big/\left|\Delta\boldsymbol{r}_H^n\right|$). The INS- and RTK-related errors are subsequently analyzed in detail prior to the evaluation of the alignment accuracy.

Traditionally, the INS-indicated position is computed through the INS mechanization procedure, and the practical algorithm and implementation have been provided in many textbooks, such as (Savage 2007). Through this approach, the positioning solution tends to drift significantly even in over a short period due to the large biases in both the accelerometer and gyro measurements, which would degrade the accuracy of $\Delta\hat{\boldsymbol{r}}_H^n$ and, in turn, influence the initial heading estimation in the proposed method.

In the present work, we calculate the INS-indicated position in a new way to largely prevent its accuracy drift. It is clear that the pole movement in the alignment phase is a typical motion of a rigid body with a fixed point. The INS-indicated APC trajectory could be straightforwardly determined based on the gyro-derived pole attitude and the constant pole length. The INS-indicated APC position is calculated by (2) in its discrete time form as:

$$\boldsymbol{r}_I^n(t_k) = \boldsymbol{r}_T^n - \mathbf{C}_b^n(t_k)\boldsymbol{l}^b \tag{17}$$

where t_k denotes the discrete time instant, $\boldsymbol{r}_T^n$ and $\boldsymbol{l}^b$ are known in advance, and the attitude is updated from the start time epoch by integrating the gyro measurement using the direction cosine matrix chain rules (Savage 1998), as:

$$\mathbf{C}_{b(k)}^{n(k)} \approx \mathbf{C}_{b(k)}^{n} = \mathbf{C}_{b(k-1)}^{n}\mathbf{C}_{b(k)}^{b(k-1)} \tag{18}$$

$$\mathbf{C}_{b(k)}^{b(k-1)} = \mathbf{I} + \frac{\sin\phi_k}{\phi_k}\boldsymbol{\phi}_k + \frac{1-\cos\phi_k}{\phi_k^2}\left(\boldsymbol{\phi}_k\times\right)\left(\boldsymbol{\phi}_k\times\right) \tag{19}$$

$$\boldsymbol{\phi}_k = \Delta\boldsymbol{\theta}_k + \frac{1}{12}\Delta\boldsymbol{\theta}_{k-1}\times\Delta\boldsymbol{\theta}_k \tag{20}$$

where k denotes the time instant t_k, $\boldsymbol{\phi}_k$ is the rotation vector defining the frame b_k attitude relative to frame b_{k-1} at time t_k, ϕ_k is the length of $\boldsymbol{\phi}_k$, and $\Delta\boldsymbol{\theta}_{k-1}$ and $\Delta\boldsymbol{\theta}_k$ are the incremental angle measurements of the IMU at t_{k-1} and t_k, respectively. $\mathbf{I}$ is the 3-by-3 identity matrix, and $\left(\boldsymbol{\phi}_k\times\right)$ is the cross-product (skew-symmetric) form of vector $\boldsymbol{\phi}_k$.

In this new approach, only gyro measurements are used in computing the INS-indicated position or trajectory, as expressed by (17). Hence, $\Delta\hat{\boldsymbol{r}}_H^n$ is free of any accelerometer-associated errors and only affected by the gyro attitude drift error. Therefore, compared to the traditional strapdown INS mechanization approach, the new method is expected to greatly reduce the positioning errors. The errors in the gyro-derived attitude change $\hat{\mathbf{C}}_{b(t)}^{b(0)}$ mainly originate from random noise, gyro bias, scale factor error, and cross-coupling errors. The impacts of the gyro bias and random noise on the attitude do not exceed 0.01°, which are both negligibly small as analyzed in the appendix. Therefore, we focus on analyzing the impacts of the scale factor and cross-coupling errors below.

The total angular rate measurement errors due to the scale factor and cross-coupling errors are calculated as:

$$\delta\omega_z = s_{g,z}\omega_z^b + m_{g,zx}\omega_x^b + m_{g,zy}\omega_y^b \tag{21}$$

where $m_{g,zx}$ and $m_{g,zy}$ are coefficients of the x-axis and y-axis angular rates, respectively, sensed by the z-axis gyro. ω_x^b, ω_y^b and ω_z^b are the x-axis, y-axis and z-axis angular rates, respectively, $s_{g,z}$ refers to the scale factor error of the z-axis gyro, and $\delta\omega_z$ is the total angular rate measurement error of the z-axis gyro, i.e., heading gyro. Dollar-level MEMS IMUs typically exhibit a scale factor and cross-coupling errors of 0.5% according to the datasheets. Such dynamic errors cannot be estimated in real-time case within a short time due to the lack of observability under the alignment conditions of tilted RTKs. Therefore, we have to accept these errors in full magnitude.

We can necessarily derive from (21) that the cross-coupling error produces a heading error proportional to the tilt angle change, i.e., the flare angle when shifting the pole, and the scale factor $s_{g,z}$ produces a heading error proportional to the heading change during the alignment motion. Suppose that we typically waggle the pole over an angle of 60°, i.e., tilted tilt of 30°, and change the heading 20° (empirical value from experiments). Hence, the cross-coupling produces a heading error of $60\times0.5\% = 0.3°$, and the heading error due to the z-axis gyro scale factor error is $20\times0.5\% = 0.1°$. Neglecting the correlations between the cross-coupling and scale factor errors, the standard deviation of the resulting heading alignment error is $\sqrt{0.1^2 + 0.3^2} = 0.32°$. Therefore,

according to the analysis above, the gyro scale factor and cross-coupling errors are the major error sources, and the gyro-derived attitude is at the accuracy level of 0.32°.

Next, we will analyze the impact of the RTK-indicated trajectory on the alignment accuracy. The RTK position is normally considered to have an accuracy of 2 cm (RMS) in the horizontal direction. But such absolute accuracy measure is not what we need here, because we are dealing with the delta position, i.e., position increment vector, which is only affected by the relative accuracy in time domain during the short alignment time (i.e., 2 s). According to the previous study by Niu et al. (2014), GNSS positioning errors show time correlation and the relative errors between two RTK epochs with 2-s interval are about 3 mm (RMS) in the horizontal direction. This indicates that the delta RTK position measurement errors should also be accurate to the level of 3 mm (RMS). Considering the impact of the antenna tilt during the operation which degrades the GNSS signal elevation condition (Luo et al. 2018), the delta RTK position, i.e., position increment vector, measurement error is empirically scaled up to 10 mm (RMS). Therefore, the influence of the delta RTK position error to the heading alignment should be at the level of $0.01(m)/\,0.02(m) \approx 0.005rad \approx 0.28°$, where 2.0 (m) is forgotten the trajectory length in the horizontal direction when the pole is tilted to 30° and 0.01 m is the delta RTK position error.

Therefore, combining both the INS-indicated and RTK-indicated errors, the standard deviation of the heading alignment error is about $\sigma_{\dot{\psi}} = \sqrt{\sigma_{\psi,I}^2 + \sigma_{\psi,G}^2} = \sqrt{0.32^2 + 0.28^2} = 0.42°$.

Experiment and results

An experiment was conducted to validate the feasibility and evaluate the performance of the proposed heading initial alignment algorithm. The field test was conducted under open-sky conditions on a campus playground at Wuhan University. The choice of favorable observation condition is reasonable because tilted RTK receivers are generally initialized under open-sky conditions. Datasets were collected using a MEMS GNSS/INS integrated system, two geodetic GNSS receivers and a quasi-tactical grade GNSS/INS integrated system to provide a heading reference. The MEMS GNSS/INS integrated system, named the INS-Probe, was developed by the Navigation Group of the GNSS Research Center at Wuhan University. In this a system, Invensense ICM-20602, a dollar-level IMU chip, is integrated and synchronized with a built-in GNSS receiver card from u-blox. Two high-quality GNSS receivers from Trimble Inc. were employed to perform RTK positioning. The independent reference system, named M39, is a GNSS/INS integrated system from Wuhan MAP Space Time Navigation Technology, Ltd., China. It integrates a quasi-tactical grade IMU, ADIS16488 from Analog Devices Inc. (ADI), with a NovAtel OEM-7 receiver card from NovAtel Inc., Canada. Figure 4 shows a photograph of the experimental setup and illustrates how the sensors were connected. An aluminum plate was fixed on the top end of a Trimble carbon fiber pole, the INS-Probe, and M39, and the antennas were firmly mounted on the plate. Table 1 lists the main specifications of the equipment, and more information on the equipment is listed below:

- IMU: Invensense ICM-20602, a dollar-level MEMS IMU.
- GNSS rover receiver: a Trimble NetR9 GNSS receiver.
- GNSS base station receiver: a Trimble NetR9 GNSS reference receiver.
- Independent heading reference: M39, which has been proven to provide an independent integrated heading reference accurate to 0.1° in postprocessing.
- GNSS antenna: NovAtel GPS-702-GGL, and a u-blox ANN-MB multiband GNSS antenna.

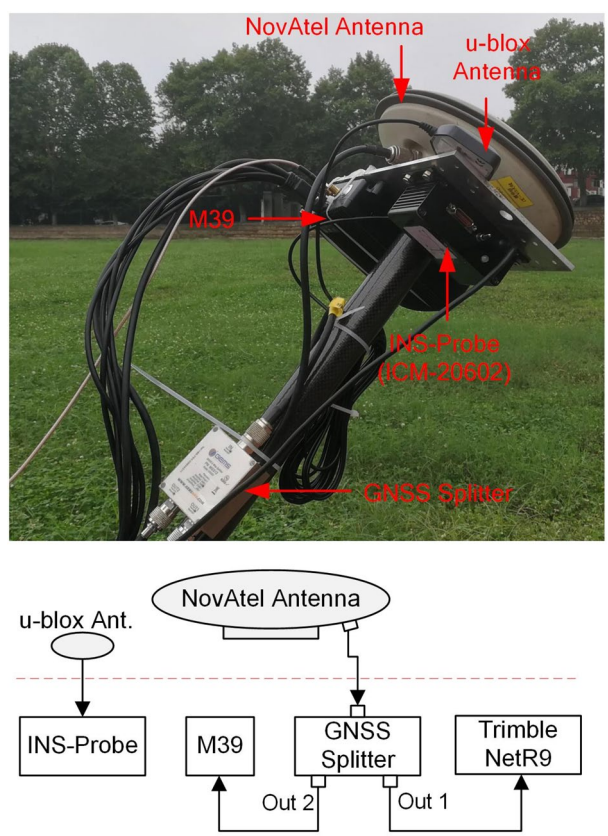

Fig. 4 Photograph of the experimental setup. The rover Trimble Net R9 receiver is not shown in the upper photograph

Table 1 Specifications of the key equipment

M39	Data rate: IMU 200 Hz, GNSS 1 Hz Gyroscope performance: Input range ± 450 deg/s In-run bias stability 6 deg/h Angular random walk 0.3 deg/ $\sqrt{h}$ Accelerometer performance Range ± 18 g In-run bias stability 0.1 mg Velocity random walk 0.029 m/s/ $\sqrt{h}$
ICM-20602	Size: 3 mm × 3 mm × 0.75 mm Data rate: 200 Hz Gyroscope noise: 0.24 deg/ $\sqrt{h}$ Accelerometer noise: 100 μg/ $\sqrt{Hz}$
Trimble NetR9	Sampling rate: up to 50 Hz (configurable) RTK surveying performance: Horizontal 8 mm + 1 ppm RMS Vertical 15 mm +1 ppm RMS
GPS-702-GGL	Signals tracked: GPS L1/L2, GLONASS L1/L2, L-band, BDS B1, Galileo E1

The baseline between the rover receiver and base station is about 1000 m. The offset from the rover GNSS APC to the pole tip is 2.1 m. Both the INS-Probe and M39 recorded synchronized IMU data at 200 Hz. The two Trimble receivers recorded both 1 Hz and 10 Hz raw GNSS data simultaneously to allow for carrier phase-based differential GNSS positioning processing. To fully test the proposed algorithm and draw a solid statistical conclusion on the alignment accuracy, datasets were collected during four independent sessions according to the following procedure:

(1) Fix the pole tip at a given point on the ground, maintain the pole stationary over the first 3 s to determine the initial roll and pitch angles and the initial gyro biases, and shift the pole over a tilt angle α over 2 to 3 s to finish one alignment recording.
(2) Repeat step 1 about 200 times to obtain multiple alignment samples under the given tilt angle α.
(3) Repeat steps 1 and 2 for $\alpha = 15°$, 30°, and 45° to evaluate the performance under the different tilt angles.
(4) Restart the sensors and repeat steps 1 to 3 to collect data in four different sessions.

Notably, the position of the point on the ground could be determined using the traditional RTK mode by keeping the pole vertically and remaining stationary for a while, e.g., 5 s, at the beginning of step 1. Step 2 is the complete alignment operation needed for the proposed algorithm, which we call the alignment sample. Therefore, each session contains about 600 alignment samples (200 samples under each tilt angle), and the whole experiment contains about 2400 alignment samples.

Data processing and result analysis

The IMU and GNSS data from M39 were postprocessed to provide an independent heading reference that was proven accurate to 0.1°. The GNSS data from the base station and rover receivers were processed in the carrier phase-based differential positioning mode in forward filtering just as the RTK positioning works to provide 1 Hz and 10 Hz RTK positioning solutions. It should be noted that although we validate the algorithm by processing the collected data, the algorithm could be implemented in a real-time case. The IMU ICM-20602 data and resulting GNSS RTK position were processed one by one for the 2400 alignment samples using the proposed algorithm. Figure 5 shows a typical sample of the 1 Hz RTK positions considered in the alignment data processing, and there are three position increment vectors available, i.e., vectors $\overrightarrow{ab}$, $\overrightarrow{bc}$ and $\overrightarrow{ac}$, if the alignment motion is finished in 3 s. In the following analysis, we choose the longest one (i.e., $\overrightarrow{ac}$) to calculate the initial heading angle. Finally, the heading alignment results were compared to the reference heading from M39 to obtain the initial heading alignment errors.

Figure 6 depicts the heading alignment errors of the 2400 alignment samples in the four different sessions, with each session containing approximately 600 alignment samples. The lower panel of this figure depicts the tilt angles for each alignment sample. It is clear from this plot that in each session, the pole was tilted over three different angles, and the alignment motion was repeated. The upper panel shows that the initial heading alignment errors mostly range from

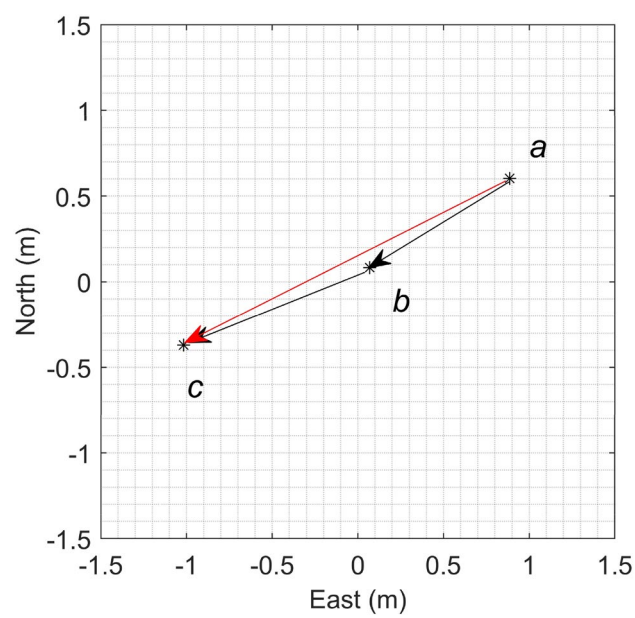

Fig. 5 Sample of the RTK positions (1 Hz) for alignment data processing

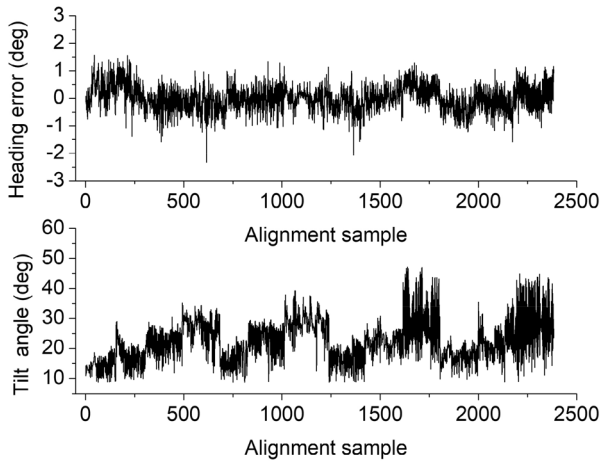

Fig. 6 Initial heading alignment errors (top) and tilt angles (bottom) in the four different sessions

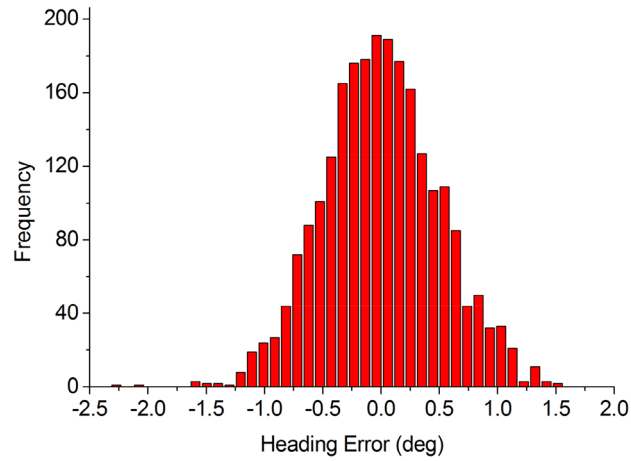

Fig. 7 Distribution of the initial heading errors

approximately −1 to 1°. Figure 7 depicts the distribution of the heading alignment errors, which is approximately a Gaussian distribution. The RMS of the heading alignment errors is 0.49°, and 98.2% of the samples are within 1.15°. Luo et al. (2018) demonstrated that the Leica tilted RTK receiver GS18 can achieve a 2° accuracy within 10 s under a sufficient motion. From this comparison, we can conclude that the proposed alignment method is sufficiently accurate for the tilted RTK receiver and is a much more time-efficient method than the conventional alignment methods.

Upon closer observation of Fig. 6, we find that the alignment errors correlate to the tilt angles to a certain extent. For example, for the sample number ranges of 1620–1800 and 2200–2400 in this figure, where the tilt angles are large, the initial heading errors seem to increase. To investigate this correlation, we sort the heading alignment error sequences in ascending tilt angle order and plot them in Fig. 8. The heading alignment error increases by 0.4° when the tilt angle exceeds 30° and reaches approximately 50°. The reasons for this phenomenon may be twofold: (1) The RTK positioning accuracy decreases with increasing tilt angle due to the smaller elevation angle of the incoming GNSS signal, as explained in the section on methodology. In our receivers, no additional advanced signal tracking technology was adopted. (2) As the tilt angle increases, the impact of the uncompensated gyro scale factor and cross-coupling errors becomes more notable.

The heading alignment results are also assessed in another indirect approach: The instantaneous tilted RTK position accuracy is assessed immediately after completing the heading alignment. About 380 instantaneous tilted RTK surveys were performed with the tilt angles ranging from 15° to 45°. The corresponding tilt-compensated pole tip position errors are plotted in Fig. 9. This indicates that after tilt

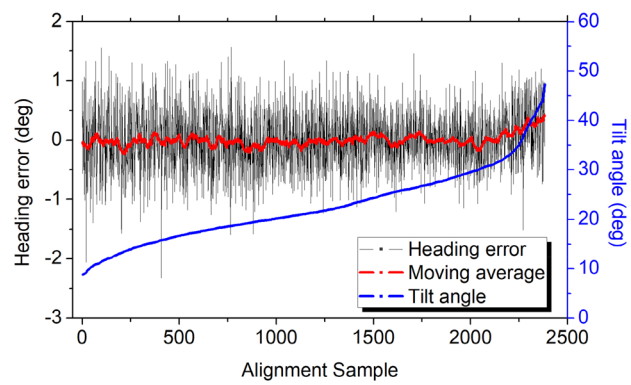

Fig. 8 Correlation between the heading alignment error and tilt angles

compensation, the instantaneous positioning solution attains a similar accuracy to that of traditional RTK surveying. This accuracy is also similar to the results from the commercial tilted RTK receiver Leica GS18, where the mean value and standard deviation are 0.012 m and 0.006 m, respectively. This indicates that the proposed heading alignment method is accurate enough for tilted RTK surveys. The height error seems noisier in this figure because the RTK height achieves a lower accuracy in the vertical direction than in the horizontal direction.

As mentioned at the beginning of the data processing description, only the longest RTK position increment vector is used as the RTK-indicated trajectory when calculating the initial heading. We can also use the multiple vectors available to determine the initial heading using the LS estimator. We use the 10 Hz RTK position for the same alignment processing approach, which means that there would be 20–30 position incremental vectors available. Figure 10 shows the

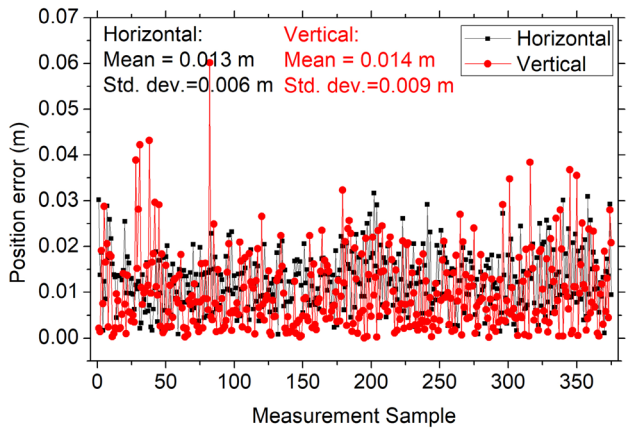

Fig. 9 Instantaneous tilted RTK position errors

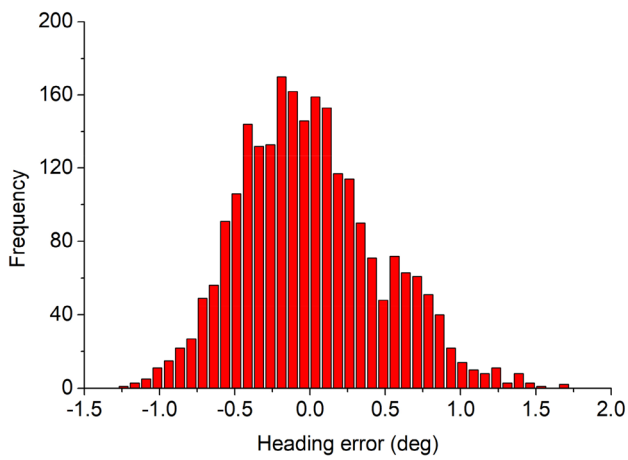

Fig. 10 Distribution of the initial heading errors using the10 Hz RTK position data

distribution of the alignment errors using the 10 Hz RTK position and all RTK-indicated position increment vectors, in which case about 20–30 vectors are available when the alignment is finished within 2–3 s. Comparing the result to that shown in Fig. 7, there is no notable accuracy improvement by increasing the GNSS data rate and using more RTK-indicated position increment vectors. This may occur because the multiple RTK-indicated position measurements at 10 Hz within the alignment are highly correlated in the time domain. However, we notice that the alignment errors exceeding 2° disappear. This indicates that a high GNSS data rate would generate a large number of measurements that could be incorporated in the initial heading determination. Thus, with redundant measurements, we have a chance to reject outliers in the measurement to make the results more stable. Also, a high GNSS data rate may accelerate the alignment procedure without requiring a period of 2–3 s.

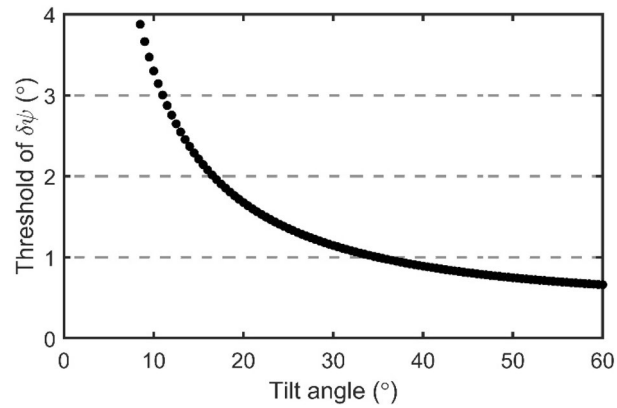

Fig. 11 Heading angle accuracy requirement versus tilt angle by assuming the pole length $l = 2.0$ m

Conclusion

Tilted RTK surveys face the challenge of rapid and accurate initial heading alignment of the built-in MEMS INS. In this research, we address this issue under a new approach, where the initial INS heading is determined from the angle between the INS- and RTK-indicated trajectories. The alignment motion is implemented as follows: Fix the pole tip at a ground point O, tilt the pole at an arbitrary position A, and then shift it to another position B to complete the alignment movement. Instead of calculating the INS-indicated trajectory through conventional INS mechanization, we compute the relative trajectory using only the gyro-derived attitude and pole length by treating the pole movement as that of a rigid body with a fixed point. Therefore, such an INS-indicated trajectory has a much smaller drift error since it is free of any accelerometer errors, which greatly enhances the alignment accuracy. The field test results reveal that the initial heading could be accurately determined to 1.15° at a 98.2% confidence level within only 2–3 s. These results proved that the proposed heading initial alignment method could meet the accuracy requirements of tilted RTK receivers, and it is a more time-efficient approach than any published methods. The proposed method is expected to speed up the initialization procedure and greatly improve the tilted RTK performance. Another potential application of this method is to enhance or recover the heading once the GNSS/INS integrated heading solutions degrade to an unsatisfactory level at the tilted RTK surveying phase, which is another technical issue for tilted RTK receivers, as mentioned in the introduction.

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Data Availability The raw GNSS and IMU data are available by contacting the authors (Qijin Chen, chenqijin@whu.edu.cn; or Xiaoji Niu, xjniu@whu.edu.cn).

Appendix

In this appendix, we address the question of how accurate the attitude angles should be for the tilt compensation when 2 cm compensation accuracy is required in the tilted RTK surveying. We will illustrate that the roll and pitch angles can be initialized with sufficient accuracy quickly even for low-cost MEMS IMU. Then, the derivation of the initial heading accuracy requirement is sketched.

Roll and Pitch Angle Initialization

When the INS remains stationary with respect to the earth, the roll and pitch angles, as defined in the methodology section, can be determined through the accelerometer leveling process in a few seconds, even for a low-cost MEMS IMU. The resulting roll and pitch errors may be expressed as follows for the particular situation in which the body frame is nominally aligned with the local geographic frame (Titterton and Weston 2004):

$$\delta\theta = -\frac{\delta f_x}{g} \tag{22}$$

where $\delta\phi$ and $\delta\theta$ denote the errors in the initial roll and pitch angles, respectively. δf_x and δf_y are the accelerometer measurement error in x-axis and y-axis, respectively. g is the magnitude of the local gravitational acceleration.

Assume the typical MEMS accelerometers are accurate to 1000 mGal, then a 0.06° roll and pitch accuracy can be achieved, which is sufficiently accurate for the RTK tilt compensation.

Heading Accuracy Requirement Analysis

Here, we analyze the heading accuracy requirement for the tilt compensation. Recall Eq. (1), the tilt compensation from GNSS antenna phase center to the pole tip in the n-frame is given by:

$$\boldsymbol{l}^n = \mathbf{C}_b^n \boldsymbol{l}^b \tag{23}$$

where $\boldsymbol{l}^n$ is the lever-arm vector resolved in the n-frame, i.e., the tilt compensation component; The lever-arm vector in the b-frame is $\boldsymbol{l}^b = \begin{bmatrix} 0 & 0 & l \end{bmatrix}^T$, where l denotes the length between the GNSS antenna phase center and the pole tip.

The direction cosine matrix in terms of Euler angles, as defined in the methodology section, is given by Savage (2007):

$$\mathbf{C}_b^n = \begin{bmatrix} c\theta c\psi & s\phi s\theta c\psi - c\phi s\psi & c\phi s\theta c\psi + s\phi s\psi \\ c\theta s\psi & s\phi s\theta s\psi + c\phi c\psi & c\phi s\theta s\psi - s\phi c\psi \\ -s\theta & s\phi c\theta & c\phi c\theta \end{bmatrix} \tag{24}$$

where c and s refer to the cosine and sine operators, respectively. The computed direction cosine matrix which contains errors due to the errors in roll, pitch and heading Euler angles, is written as follows (Titterton and Weston 2004):

$$\hat{\mathbf{C}}_b^n = [\mathbf{I} - \boldsymbol{\Psi}]\mathbf{C}_b^n \tag{25}$$

where $\mathbf{I}$ is the 3-by-3 identity matrix and $\boldsymbol{\Psi}$ is given by:

$$\boldsymbol{\Psi} = \begin{bmatrix} 0 & -\delta\psi & \delta\theta \\ \delta\psi & 0 & -\delta\phi \\ -\delta\theta & \delta\phi & 0 \end{bmatrix} \tag{26}$$

where the elements, $\delta\phi$, $\delta\theta$, and $\delta\psi$ correspond to the errors in roll, pitch and heading Euler angles, respectively. The tilt compensation error, denoted by $\delta\boldsymbol{l}^n$, due to the errors in the computed attitude matrix is given by:

$$\delta\boldsymbol{l}^n = \hat{\mathbf{C}}_b^n \boldsymbol{l}^b - \mathbf{C}_b^n \boldsymbol{l}^b \tag{27}$$

Substituting (25) into (27) yields

$$\delta\boldsymbol{l}^n = -\boldsymbol{\Psi}\mathbf{C}_b^n \boldsymbol{l}^b \tag{28}$$

Substituting (24) and (26) into (28) yields the three elements of $\delta\boldsymbol{l}^n$ as:

$$\begin{cases} \delta l_N = \delta\psi \cdot l(c\phi s\theta s\psi - s\phi c\psi) - \delta\theta \cdot l(c\phi c\theta) \\ \delta l_E = -\delta\psi \cdot l(c\phi s\theta c\psi + s\phi s\psi) + \delta\phi \cdot l(c\phi c\theta) \\ \delta l_D = -\delta\phi \cdot l(c\phi s\theta s\psi - s\phi c\psi) + \delta\theta \cdot l(c\phi s\theta c\psi + s\phi s\psi) \end{cases} \tag{29}$$

Equation (29) shows that the heading angle error, i.e., $\delta\psi$, only influences the tilt compensation in the horizontal direction. The roll and pitch angle errors, i.e., $\delta\phi$ and $\delta\theta$, have an effect on both the horizontal and vertical directions. Even for MEMS IMU, the roll and pitch angles can be computed accurately corresponding to the previous analysis. The components of the tilt compensation errors induced by $\delta\psi$, are given by:

$$\begin{cases} \delta l_{\delta\psi,N} = \delta\psi \cdot l(c\phi s\theta s\psi - s\phi c\psi) \\ \delta l_{\delta\psi,E} = -\delta\psi \cdot l(c\phi s\theta c\psi + s\phi s\psi) \end{cases} \tag{30}$$

where $\delta l_{\delta\psi,N}$ and $\delta l_{\delta\psi,E}$ are the north and east component of the $\delta\psi$-induced tilt compensation errors in the horizontal direction. Thus, the magnitude of the combined $\delta\psi$-induced

tilt compensation errors in the horizontal direction is computed as follows:

$$\delta l_{\delta\psi,H} = \sqrt{\delta l_{\delta\psi,N}^2 + \delta l_{\delta\psi,E}^2} = \delta\psi \cdot l\sqrt{1 - c^2\phi c^2\theta} \tag{31}$$

Substituting (24) into (23), the third element of the vector $\boldsymbol{l}^n$, i.e., the vertical component of the tilt compensation, is obtained as:

$$l_D^n = l\cos\phi\cos\theta \tag{32}$$

According to (Luo et al. 2018), the tilt compensation relates to the tilt angle (denoted by α), which is defined as the angle between the local zenith and the pole, as depicted in Fig. 1. We can readily write the vertical component of the tilt compensation with respect to the tilt angle α, as:

$$l_D^n = l\cos\alpha \tag{33}$$

Comparing (32) and (33) yields

$$\cos\alpha = \cos\phi\cos\theta \tag{34}$$

Substituting (34) into (31) yields

$$\delta l_{\delta\psi,H} = \delta\psi \cdot l\sin\alpha \ \ (0° \le \alpha < 90°) \tag{35}$$

The above equation demonstrates that tilt compensation errors are introduced as a result of the heading angle error, i.e., $\delta\psi$, the length of the offset between GNSS antenna phase center to the pole tip, and the tilt angles. It should be noted that in (31) and (35), we denote $\delta l_{\delta\psi,H}$ as the magnitude of the horizontal compensation error by ignoring the sign of the quantity. Figure 11 is a plot of the thresholdof the heading angle error versus the tilt angles when 2 cm tilt compensation accuracy should be assured for a 2-m lever arm. For example, for a 2 m pole, a 1.15° accuracy is required when the pole is tilted 30° and 2.0° accuracy is required when the pole is tilted 17°.

Impacts of the Gyro Bias and Random Noise on the Heading Alignment

The random noise in the gyro angular rate measurements is integrated to produce an attitude random walk error. The standard deviation of a random walk process is proportional to the square root of the integration time (Groves 2013). It is common for manufacturers to specify the random noise using the angular random walk (ARW) or power spectral density (PSD). The ARW for the random noise of low-cost gyro chips used in tilted RKT receivers is typically $0.3°/\sqrt{h}$, please refer to the specifications of ICM-20602 listed in Table 1 in the experimental description. This means that after 1-hour integration, the standard deviation of the attitude error is 0.3°. As the proposed heading alignment can be completed in 2 s, the attitude drift produced by random noise will be $0.3 \cdot \sqrt{2/3600} \approx 0.01°$. Therefore, the impact of the random noise on the gyro-derived attitude is negligibly small.

The fixed gyro bias may comprise the run-to-run variation plus the residual fixed bias remaining after sensor calibration. A fixed gyro bias produces an approximately linear attitude drift within a short time, as follows:

$$\delta\beta = b_g t \tag{36}$$

where $\delta\beta$ is the attitude drift, and b_g is the fixed gyro bias. The low-cost gyro chips employed in tilted RTK receivers may exhibit large run-to-run gyro biases up to 1.0°/s. In practice, this large fixed gyro bias could be initially estimated by averaging the gyro measurements when the IMU remains stationary. Please note that the earth rotation rate, i.e., the actual input of the gyros when stationary which is about 15°/h, is ignored here and therefore is left as an error source of such gyro bias estimation. Another error source is the impact of the gyro noise to the averaging value of the gyro measurement. Averaging the stationary gyro measurements is a process to filtering out the random noise. As long as the random noise could be averaged out to a sufficient level, the run-to-to bias could be estimated to a certain extent. The standard deviation of the average gyro angular rate measurement varies inversely proportionate to the square root of the averaging time. If the tilted RTK receiver remains stationary for 3 s during roll and pitch initialization and the ARW is $0.3°/\sqrt{h}$, as mentioned above, the standard deviation of the average gyro measurement error is $0.3/\sqrt{3/3600} \approx 10.4°/h$. Considering this residual impact of the random noise plus the ignored earth rotation rate (15°/h), the run-to-run gyro bias could be estimated to the level of $\sqrt{10.4^2 + 15.0^2} = 18.2°/\text{h}$. After compensating the estimated bias to the gyro measurement, the residual run-to-run bias is reduced to a magnitude of approximately 18.2°/h. Therefore, the attitude drift induced by the gyro bias does not exceed 0.01°, please refer to (36), when the alignment is accomplished within 2 s, which is negligibly small.

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